

Academic integrity declaration

By submitting work for assessment, I hereby declare that I understand the University's policy on academic integrity and the rules of the assignment as set by the subject coordinator regarding the use of artificial intelligence software, including editing and translation tools.

I declare that the submission is original and solely my work, and that I have not been assisted by another person (collusion) apart from where the submitted work is for a designated collaborative task, in which case the individual contributions are indicated. All sources and tools used have been appropriately cited or acknowledged, and any copied or modified code has been declared using the methods indicated by my subject coordinator.

System Project: Final Assessment 35 Points Possible



2026/11/13

Attempt 1



In Progress

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Unlimited Attempts Allowed

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Marking Scheme for: System Project: Technical Details

The five standards below assess your ability to understand, investigate, justify, and communicate the technical details of the System Project:

- **These standards are all assessed during the final oral assessment for the project project.**
- **Each standard is worth 7% of the overall subject grade.**

The descriptions below define the Meets level for each standard. As per the [grading approach page \(https://canvas.lms.unimelb.edu.au/courses/238681/pages/grading-approach-for-elen90095-2026-sm2\)](https://canvas.lms.unimelb.edu.au/courses/238681/pages/grading-approach-for-elen90095-2026-sm2), the generic descriptions of Exceeds, Approaching, and Not yet demonstrated are taken relative to the Meets level.

Demonstrating Engineering justification and judgement is an inherent part of Standards 1-4. In other words, demonstrating the technical details of Standards 1-4

requires more than describing the final result. Where relevant, your assessment evidence should explain and justify the engineering process that you carried out. This can include: alternatives considered; assumptions made; constraints encountered; trade-offs evaluated; decisions made; and limitations identified.

Standard 5 about visual communication is a standalone standard because assessment of visual communication quality can be sufficiently disambiguated from Standards 1-4.

Due to the open-ended nature of the System Project, there is some flexibility in the bullet points that describe what your assessment evidence should provide. You should focus on conveying the most relevant evidence for the technical work that you undertook.

Standard 1: System-level thinking and understanding

My assessment evidence demonstrates that I can formulate and explain how a subsystem fits within the overall system and project, including its function, interfaces, design goals, assumptions, and constraints.

To meet this standard, the assessment evidence should provide:

- A clear explanation of the subsystem's purpose and how it contributes to the overall system and project goals.
- Description of the important inputs, outputs, interfaces, dependencies, requirements, assumptions, and constraints that shape how the subsystem operates within the wider system.
- Technical reasoning that connects subsystem design decisions and behaviour to the performance and behaviour of the overall system.

Standard 2: Theory, algorithms, and technical details

My assessment evidence demonstrates that I can synthesise and explain the theory, algorithms, equations, and relevant technical details at the core of a subsystem I have worked on.

To meet this standard, the assessment evidence should provide:

- A clear explanation of the important theory, algorithms, equations, and technical concepts underlying the subsystem.
- Definitions and interpretation of important quantities, parameters, assumptions, and relationships.
- Explanation of how the theory and/or algorithms connect to the implemented subsystem and its observed behaviour.

Standard 3: Simulation-based investigation

My assessment evidence demonstrates that I can use simulation to develop and investigate technical approaches that are relevant to real-world deployment of the system.

To meet this standard, the assessment evidence should provide:

- A clear engineering question or design objective that the simulation is being used to investigate.
- Appropriate use of simulation to develop, tune, compare, or evaluate algorithms, controllers, models, policies, or other technical approaches.
- Meaningful simulation evidence and comparisons that support interpretation of system behaviour, sensitivities, operating conditions, and/or failure cases.
- Justification of what can reasonably be concluded from the simulation, including important assumptions and limitations for real-world deployment.

Note: Simulation can be interpreted broadly and is NOT limited to the provided Gymnasium simulation environment. Simulation can include any appropriate computational model or virtual representation used to investigate aspects of subsystem and/or overall-system behaviour.

Standard 4: Real-world experimental investigation

My assessment evidence demonstrates that I can conduct controlled experiments on the physical system to investigate behaviour relevant to real-world deployment, performance, and integration.

To meet this standard, the assessment evidence should provide:

- A clear engineering question together with an experimental method designed to investigate it.
- Appropriate control or measurement of influential variables, with experimental conditions described sufficiently to interpret the results.
- Meaningful experimental results and justified interpretation of what they demonstrate, including limitations and implications for engineering decisions.

Note: Experimental investigations can take many forms, including: parameter identification; characterising subsystem capabilities; checking limitations; robustness testing; comparison with simulation; and/or validation of whole-system integration.

Standard 5: Visual communication of technical details

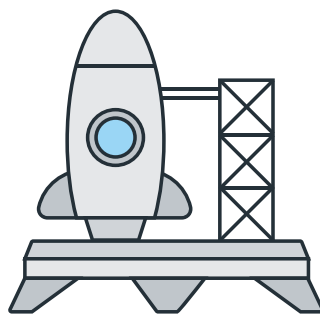
My assessment evidence demonstrates that I can use appropriate visual forms of communication to make technical details, reasoning, and evidence easier to understand.

To meet this standard, the assessment evidence should provide:

- Visual forms that are appropriate and effective for conveying the respective structures, relationships, behaviours, comparisons, and/or results.
- Appropriate labels, annotations, scales, legends, units, and other information needed to interpret the visual evidence accurately.
- Integration of visual and verbal explanation so that the visuals actively support the technical reasoning rather than merely decorate the presentation.

Note: visual communication appropriate for this standard can take many different forms, such as, sketches, system diagrams, equations, plots, tables, annotated images, animations, or video.

Choose a submission type.



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